

UNIVERSITY PARTNER



Project and Professionalism

EDITEASE: A Video Organizing Platform

Name: Ashreen Dangol

Student ID: 2407774

Supervisor: Kaushal Kishor Mishra

Word Count: 13640

Submitted on: 3/4/26

Table of Contents

Table of Contents	2
Table of Figures	4
Abstract	6
Introduction	8
1.1 Background of the Study	8
1.2 Problem Statement	9
1.3 Aim of the Project	10
1.4 Objectives of the Project	10
1.5 Artefact Overview	12
1.6 Academic Question	13
1.7 Scope and Limitations	13
1.8 Structure of the Report	14
Literature Review	15
2.1 Literature Review Introduction.	15
2.2 Multimedia Data and Video Content Analysis.	16
2.3 Scene Detection and Video Segmentation.	17
2.4 Video Classification and Content Understanding.	18
2.5 Video Summarization	19
2.6 Video-based Emotion Recognition.	20
2.7 Multimedia Retrieval Systems and Metadata.	21
2.8 Human-in-the-Loop Systems	22
2.9 Literature Findings.	22
Project Methodology	23
3.1 Introduction to the Methodology.	23
3.2 Development Approach	24
3.3 System Development Phases	25
3.4 System Architecture Overview.	27
3.5 Data Processing Workflow	28
Technologies and Tools Used	30
4.1 Introduction	30
4.3 Video Processing Libraries	31
4.4 Database Technology	31
4.5 Backend Framework	32
4.6 User Interface Framework	32
4.7 Version Control and Development Environment.	33

System Architecture and Artefact Design.	34
Introduction to the Design of the Artefact.	34
5.2 System Architecture Overview.	35
5.3 Subsystem Decomposition	37
5.4 Video Input and Processing Subsystem.	37
5.5 Scene Detection Subsystem	38
5.6 Thumbnail Subsystem Extraction.	39
5.8 Subsystem of Metadata Generation.	40
5.8 Database Design	41
5.9 Application Programming Interface (API)	43
5.10 User Interface Design	43
5.11 Data Flow within the System	44
5.12 Summary of Artefact Design	45
6 System Implementation	47
6.1 System Implementation Introduction.	47
6.2 Implementation of Video Processing Pipeline.	47
6.3 Implementation of Metadata Generation.	50
6.4 Database Implementation	50
6.5 API Implementation	52
6.6 Implementation of the User Interface.	53
System Evaluation and Testing.	54
7.1 Introduction to System Evaluation.	54
7.2 Functional Testing	55
7.3 Usability Evaluation	55
7.4 Performance Factors.	56
7.5 Limitations Discovered in the course of testing.	56
Critical Assessment of the Project.	57
8.1 Introduction to the Critical Evaluation.	57
8.2 Assessment of Project Objectives	58
8.3 Evaluation of System Design	59
8.5 Limitations of the System	61
8.6 Future Improvements	62
Project Management Evidence.	63
9.1 Project Planning	63
9.2 Development Timeline	63
9.3. Meetings and Tracking of progress with Supervisors.	64
9.4 Risk Management	65

10. References	66
Project meeting logs	68
Logsheet1	68
Logsheet2	69
Logsheet3	70
Logsheet4	71
Logsheet5	72
Logsheet6	73
Logsheet7	74
Logsheet8	75
Logsheet9	76
Logsheet10	77
Gantt chart or development timeline	78

Abstract

The explosion in the digital video content has led to the creation of huge amounts of raw content that cannot be utilized in post-production processes before it has been reviewed and catalogued manually. Providing editing and content creators with the issue of working with long recordings in order to find certain elements like testimonials, audience reactions, establishing shots, or supporting B-roll shots. It is a time consuming and inefficient manual process especially when the amount of material recorded in the various projects is at play. Consequently, the demand to have automated systems that can be used to aid the organization and access to video content in a structured and searchable format has increased.

The presented project proposes an AI-supported video management and search system called EditEase which is aimed at making the process of cataloging raw video footage easier. The system aims at converting the unstructured video files into structured data on the scene level that can be easily browsed, filtered, reviewed, and exported via a user-friendly interface. The platform works with video files to detect scenes with the help of the scene detection method, and generates a representative frame to create a thumbnail, and the metadata about the detected scene is stored in a database. The system allows the editors to find the appropriate clips much faster than in the workflows of the traditional file-based systems because video material is divided into smaller parts with descriptive metadata.

The prototype implementation incorporates various parts such as pipeline of scene detection, metadata storage layer, application programming interface (API) to query and update operations, and visual interface to browse indexed scenes. The assistive cues of scene classification and emotion inference are not final results but assistive ones, which can help in supporting human review but not substitute editorial judgment. The project focuses on a human-in-the-loop model in which users are allowed to review identified scenes, correct labels, write notes about the scene, and export selected sections to continue working in editing software.

The findings of the prototype indicate that automated scene indexing will go a long way in enhancing the usability of raw video archives. Users can search scenes using thumbnails, metadata filters, and searchable attributes instead of viewing the complete recordings. Even though the present-day implementation is a prototype with specific shortcomings, the project provides a base on which the further work on the project will take place with better classification models, more robust dataset training, and increased interface functions.

Introduction

1.1 Background of the Study

The growing availability of digital cameras, mobile phones and recording devices has greatly increased the creation of video materials in all areas of entertainment, education, marketing, journalism and social media. With the ever-increasing size of the recorded video, the issue of storage and systematizing raw footage has emerged as a burning issue to video editors and production crews. Raw recordings usually have several scenes, cuts, and events that have to be checked by hand before some meaningful clips may be extracted to be edited.

In conventional editing processes, video files are normally saved in folders under basic naming systems. Although this method offers a simplistic organization, it fails to enable the editors to find specific moments in long recordings with ease. Using a video editing example, the editor is required to manually scrub through the entire timeline to locate a specific section of the recording when he/she is required to locate a particular testimonial clip. Such process is more ineffective in case more than one recording needs to be checked or the editor is not aware clearly where a particular scene is located. Video editing software, like timeline markers and manual tagging, can be used in the organization of modern video editing software. Nevertheless, the tools remain very manualized and do not automatically index the process. Consequently, there are high possibilities of automated systems that can help in categorizing the video contents into smaller units which can be searched and retrieved more effectively. Video analysis has seen the emergence of new opportunities due to the advances in computer vision and artificial intelligence. Information can be extracted in video frames using techniques like scene boundary detection, object recognition, face analysis and content classification. Using these methods, one will be able to divide a video into separate scenes and identify metadata with every scene. By storing this information in an organised database, the video information can be retrieved by query and filter instead of scrolling. Although these technologies have been developed, most of

the tools available are designed with single task-oriented applications, like object recognition or emotion identifiers, instead of offering an overall workflow of indexing and retrieving video scenes. It is this gap that has led to the need to have systems that integrate scene detection, metadata generation, database storage and easy to navigate browsing features into one platform.

To fulfill this requirement, the EditEase system was created to provide a platform that allows editors to handle raw video content in terms of automated indexing of scenes and searching metadata.

1.2 Problem Statement

One of the most common problems facing video production processes nowadays is dealing with large amounts of raw video footage. The recordings that the editors deal with on a regular basis are often of several minutes or hours of length, and they contain numerous scenes that are or are not likely to be included in the end product of the editing process. Detecting any useful segments in these recordings involves manually searching through the timeline in the video that is time-consuming and can be subject to human error.

It is especially challenging when it comes to locating certain scenes when one has to work with recordings of events, interviews, conferences, or documentaries. Such records are usually a combination of different types of shots such as audience reactions, parts of the speaker, environmental shots and transitional shots. In the absence of an effective indexing strategy, editors have to be directed by the memory or have to watch the footage a few times to find the required parts.

The other issue is that there is no structured metadata that is attached to the raw video files. Although files can be arranged in folders or be renamed with a descriptive name, this type of organization does not give any information regarding the internal contents of the video. As a

result, the editors cannot conduct selective searches of certain types of scenes or emotional expressions in the footage.

Even though there are commercial editing tools that offer automated scene detection capabilities, they are typically narrow-focused and lack a smooth integration with database-based retrieval systems. Also, the available solutions might not be flexible enough to offer custom workflows or experimentation in a research manner.

These drawbacks prove the necessity of such a system that will be able to process video data automatically, divide it into meaningful moments, create metadata about these moments, and store the data in the searchable form. Such a system would allow users to search and access individual clips without having to go through the whole recordings.

The EditEase project will focus on solving this issue by creating a prototype system that will be able to index video scenes and display them using an interactive browsing interface.

1.3 Aim of the Project

The main goal of the project is to design and create an AI-based video indexing and retrieval system that will allow arranging and using raw video footage effectively.

The system aims at automating the task of detecting the boundaries of scenes in video recordings and linking each scene to descriptive metadata that could be stored and queried in a database. The platform will enhance the video editing process by converting unstructured video files into structured data at the scene level.

1.4 Objectives of the Project

In order to reach the general purpose of the project, the following specific ones have been set:

- To create a scene detection pipeline that can divide the raw video files into single scenes.
- To get representative frames of each of the scenes detected to be used in the generation of thumbnails that visually depict the content present in the segment.
- To generate metadata about every scene, which contains timestamps, scene duration and classification labels.
- To create a database layout that stores metadata of scenes in a structured format that can be used to query and access.
- To develop an application programming interface (API) that will enable search, update, review, and export of scenes.
- To develop a visual interface allowing the user to navigate through indexed scenes by filtering the results using metadata and thumbnails.
- To add human-in-the-loop review systems that enable users to check or change automatically generated labels.
- To test the prototype system with regards to its capability of facilitating effective browsing and accessing of video snippets.

All these are the goals which will drive the development of the EditEase platform starting with the original idea up to the prototype implementation.

1.5 Artefact Overview

The main artefact that will be created in this project is the prototype system of EditEase. The artefact is made up of a number of interrelated elements which collectively make it possible to index and retrieve video scenes automatically.

The former element is the video processing pipeline, which processes the input video files and finds the boundaries of the scenes based on the scene detection algorithms. The detected boundaries are used by the system to extract representative frames which are used as visual thumbnails of each scene.

The second element is the metadata generation module which gathers information concerning each scene such as timestamps, duration, classification labels and reference to thumbnails. This metadata is designed in a way that it can be stored in a database.

The third one is the database layer, which contains scene metadata in a document-oriented database. Every scene is described as a separate record where all the related information is stored to be retrieved.

The fourth element is the backend application interface, which has query and update endpoints of scene data. The API enables the user interface to access the scenes by search parameters like the type of the scene, duration or the review status.

The last element is the user interface whereby the indexed scenes are displayed to the user in a graphical format. It is through this interface that users can search through scenes, filter them, examine metadata and export clips of choice.

A combination of these elements creates a prototype system that illustrates how automated video indexing and video retrieval is possible.

1.6 Academic Question

The following research question will be explored in the project:

To what extent can an AI-assisted scene indexing system enhance the scale and processing of raw video footage in comparison to conventional file-based workflows?

This query is the main force that led to the creation of EditEase. Through exploring this question, the project will establish whether automated scene segmentation and metadata indexing can minimize the amount of time and effort needed in searching and finding relevant video segments.

1.7 Scope and Limitations

The EditEase project scope majorly revolves around the actual development of a prototype system which illustrates the concept of automated indexing and retrieval of scenes. The system is used to process video files stored locally, identify the scene boundaries, extract thumbnails, create metadata and store the scene records in a database.

The prototype will serve as a research and development system and not a full-scale production system. In this regard, there are some constraints in the existing implementation.

The first weakness is that the classification of the scene is done according to heuristic rules, but not according to a trained machine learning model. Although this method is adequate to illustrate the indexing process, it might not be as accurate as the more complex deep learning models.

Another limitation relates to the handling of emotion recognition. Emotion inference is only applied when faces are detected within a scene. When no faces are present, emotion values remain undefined. This design decision helps prevent misleading interpretations but also limits the scope of emotion analysis.

Additionally, the prototype currently operates as a local application rather than a distributed or cloud-based platform. Future versions of the system may explore deployment options that support larger datasets and collaborative workflows.

Despite these limitations, the prototype provides a functional demonstration of how automated indexing can improve the organization of raw video content.

The other limitation is connected with the process of emotion recognition. Inference of emotions is done only when faces are identified in a scene. When there are no faces, the values of emotions are undefined. Such a design choice contributes to avoiding misleading interpretations, but it also restricts the extent of emotion analysis.

The prototype is also not a distributed or cloud-based application but a local application at the moment. Subsequent iterations of the system can consider the implementation possibilities of supporting more data sets and collaborative processes.

In spite of these shortcomings, the prototype offers a working example of how automated indexing can enhance the arrangement of uncoded video material.

1.8 Structure of the Report

This report will be structured into a few chapters where the development and evaluation of the EditEase system will be described.

The literature review analyzes the existing research associated with the video classification, scene detection, and automated video analysis.

The methodology chapter provides the explanation of the approach of developing the project and reasons why the adopted implementation strategy is justified.

The technologies and tools chapter explains the software frameworks and libraries that will be used to create the system.

The architecture and subsystem design of the platform are provided in the artefact design chapter.

The implementation and evaluation chapters explain the present condition of the prototype and determine its performance.

The report ends with a critical review, project management evidence and supporting appendices.

Literature Review

2.1 Literature Review Introduction.

Creation of systems that can arrange and analyze video material has grown to be a significant research field in the domains of computer vision, multimedia retrieval and artificial intelligence. Since video production in digital form keeps growing in all fields, researchers and developers have tried to find a way of automation to analyze and index video data. The body of literature on video analysis has covered varieties of methods such as segmentation of scenes, classification of videos, summarization of videos, recognition of emotions on faces, and metadata based video retrieval systems.

This literature review is aimed at analyzing the critical research areas that should be considered in developing the EditEase system. The project is based on a number of intersecting fields of multimedia computing as opposed to concentrating on one single technical technique. Such areas are scene boundary detection, automated video classification, emotion recognition in video

information and database-based multimedia content retrieval systems. Analyzing the current literature on these topics, one can gain the idea about the modern possibilities of the automated video analysis systems and determine the limitations which provide the reasons to develop the systems like EditEase.

The review thus analyzes the past studies, compares methodologies employed by various studies, and argues how such methodologies are used in the design of the proposed system. It is not merely an attempt to summarize the past work but finding the conceptual background that underpin the creation of an AI-assisted video indexing platform.

2.2 Multimedia Data and Video Content Analysis.

Video content analysis is a process that is used to retrieve meaningful information on video data by automated computational processes. Video data unlike the static images has spatial and temporal content. The spatial component visual information that exists in each frame of the image and the temporal component is the visual information that varies with time. This two-sidedness of the video data poses great challenge to the automated analysis system.

The initial studies in multimedia computing were based on the significance of the structural composition of video. A video may be defined as a series of frames that are arranged into shots, scenes and bigger story units. Scene detection and segmentation are thus important initial stages in the majority of video analysis systems. It is hard to pinpoint structural boundaries, and then it would be hard to tie the metadata to meaningful segments of the video.

The contemporary video analysis systems tend to be based on computer vision algorithms to identify visual patterns, motions and objects in frames. Such methods can enable systems to identify events, faces, gestures, and other contextual information of the video stream. Nevertheless, the video content is very complex such that the automated systems need to strike a balance between accuracy and computational efficiency.

Scholars have been more aware of the fact that the utility of video analysis is not merely based on proper detection but also on presentation and organization of the information extracted to the users. A system can easily identify objects or faces in a video, however, when the findings cannot be conveniently viewed or accessed, the practical utility of the system is constrained. This observation demonstrates the need to integrate video processing and database format and retrieval interfaces.

The EditEase project also shares this view in that it is not only concerned with the detection but with the arrangement of video scenes in searchable units. The system will be in line with the overall research direction of multimedia information retrieval by converting raw footage into organized records of scenes with related metadata.

2.3 Scene Detection and Video Segmentation.

One of the most basic tasks in video processing is scene detection. Scene detection is meant to subdivide a video into smaller parts that represent meaningful transitions in the content. Such transitions usually happen when the camera angle/location/subject/visual composition changes.

Classical methods of scene detection are based on the visual differences between the successive frames that are used to detect shot boundaries. When the gap between the frames is more than a particular threshold, the system will presume that a new scene or a shot has started. A number of algorithms have been designed to identify such boundaries which include histogram comparison, edge change ratio analysis and motion based segmentation algorithms.

Histogram comparison techniques compare the pixel intensities distributions of frames. Once the error given by the histogram of two consecutive frames is high, then it implies a potential change of scene. This method can be computationally efficient but can have trouble with the slow transitions like fades or dissolves.

More complex methods involve machine learning to identify patterns relating to transition of scenes. These models are able to learn using labeled datasets and identify transitions using a combination of visual features and not using threshold based differences. But machine learning methods need training data and computation resources that are not necessarily available on prototype systems.

Scene detection is very important in the EditEase system since it forms the basis of indexing the video content. The system is able to add metadata to small segments of videos instead of entire videos. This segmentation makes it possible to build searchable records of the scenes which can be browsed and exported in the future.

The project takes on a cut-based scene detection strategy, which aims at identifying a sudden transition between shots. This approach is suitable to a prototype implementation since it offers a good segmentation without having to train a great deal of data.

2.4 Video Classification and Content Understanding.

Video classification can be defined as the act of classifying video pieces based on the nature of activity or scene being represented. Video classification systems are developed in numerous research studies to identify some actions like walking, running, or talking. Such systems tend to be based on deep learning frameworks that can process spatial and temporal features.

Convolutional Neural Networks (CNNs) are widely employed to process spatial data of individual frames. Nonetheless, since videos have also temporal dynamics, numerous classification systems use CNNs together with sequence-based networks like Recurrent Neural Networks (RNNs) or Long Short-Term Memory (LSTM) networks. Such hybrid architectures enable the system to take into account how visual patterns change with time.

In spite of the fact that high accuracy in the use of advanced video classification models can be achieved in controlled datasets, the models also have challenges. Deep neural networks are

cumbersome to train and need extensive annotated training datasets and high-performance computing resources. Moreover, it is possible that the outcomes of such models are hard to interpret, particularly when it is used with various real-world footage.

In the case of the EditEase prototype, an easier classification strategy was used. The system does not train a complicated neural network, but rather relies on heuristic rules, which depend on visual effects like face presence and motion levels. These regulations permit scenes to be classified as either testimonials, audience reactions, and B-roll footage.

Although this rule-based model is not as advanced as deep learning models, it is a viable solution in the early stages of development. The system does not lose the opportunity to produce useful metadata that assists video content navigation, but it can be improved in the future with a machine learning model, which will be trained on the reviewed data.

2.5 Video Summarization

Another research area that is directly related to the objectives of the EditEase project is video summarization. Video summarization is aimed at producing a video with the largest amount of essential information. This is usually done by choosing some key frames or scenes that give a picture of the entire contents of the recording.

A number of methods have been suggested in order to summarize videos. Other techniques are based on the clustering of frames according to visual similarity, whereas others are based on attention mechanisms used in order to detect segments of a frame that have important activity or semantic content. Deep learning models have learned the most representative frames of the entire narrative of a video.

Although these have been made, video summarization systems tend to concentrate on making a summarized version of the video, as opposed to designing an interactive browsing mechanism.

Consequently, not a lot of summarization methods offer detailed metadata or search features of individual scenes.

The EditEase project is different to the traditional summarization systems in that it focuses on indexing and retrieval of scenes and not compression. The system does not create a condensed video but instead it breaks down the footage into segments that can be accessed separately. This will be much closer to the requirements of editors who will need specific control over the type of clips they choose to use in their end product.

2.6 Video-based Emotion Recognition.

Emotion recognition is a subfield of computer vision that is concerned with recognizing human emotional expressions on a face image or a video frame. There are numerous systems based on facial feature analysis that allows classifying emotions as happiness, sadness, anger, or surprise. These systems normally make use of machine learning models that are trained on data that includes labeled facial expressions.

The possible uses of emotion recognition are in the fields of human-computer interaction, psychology studies and multimedia analysis. Nevertheless, the validity of emotion recognition systems is a debatable issue. Emotional expression in human beings is intricate and depends on cultural, social and situational aspects. This means that automated emotion detection does not necessarily generate correct interpretations.

Ethical issues involving emotion recognition technologies have been brought into the limelight by researchers as well. The wrong interpretation of facial expressions may result in wrong assumptions regarding the emotions of a person. There is also a privacy and consent issue when it comes to the use of facial analysis.

The emotion detection in the EditEase system is done in a restrained and cautious way. The process of emotion inference is used when a face is detected in a scene and the outcomes are

handled as additional metadata and not definite labels. Users can edit and revisit the information that has been identified and make sure that the system facilitates human interpretation and not substitution.

This method is an indication that emotion recognition is to be employed with caution and that the automated prediction must in any case be checked by humans.

2.7 Multimedia Retrieval Systems and Metadata.

Metadata is very important in multimedia retrieval systems. Metadata is a structured information which defines the content, context, or attributes of digital media. Metadata may contain attributes (timestamps, category of the scene, objects detected and emotional hints) in the context of video indexing.

Through saving metadata in an organized database, search operations can be done to retrieve pertinent content using the stated requirements. As an example, a user can search by scenes that have a specific object, scenes that fall within a time frame, or a scene related to a specific emotional tone.

Multimedia management platforms heavily deploy database-based retrieval systems. The systems enable users to filter results, sort data and access certain items of immense datasets. Document-based databases like MongoDB are also quite appropriate in storing multimedia metadata since they have flexible data structure and nested fields.

The EditEase project is a database project that uses the document based database model to store scene metadata. Every detected scene is recorded as a separate record that includes the information in the form of timestamps, labels, emotion data, and thumbnail references. This hierarchy enables the retrieval of scenes in an efficient way by using search queries and filters.

2.8 Human-in-the-Loop Systems

Human-in-the-loop systems are systems that integrate automated processing and human supervision. The systems have algorithms that do initial analysis, but the users authenticate or refine the outcomes. Such a method is especially useful in areas where a completely automated interpretation can be untrustworthy.

Human-in-the-loop systems in multimedia analysis enable users to learn about automatically generated labels and fix mistakes. One can then use these corrections to create better future models through the creation of labeled datasets.

The EditEase platform is designed to follow a human-in-the-loop format by enabling users to view indexed scenes, label scenes, and add notes of context to them. Such a solution guarantees that the system is flexible and adaptable and is not over-reliant on automated predictions.

2.9 Literature Findings.

The literature reviewed in this chapter brings out several insights that are important in the development of the EditEase system.

To start with, scene detection and video segmentation are necessary processes in the arrangement of video content. Without pointing out structural boundaries, metadata can hardly be related to meaningful portions of a video.

Second, the technologies of video classification and emotion recognition can give valuable contextual information, although their results should be viewed with caution. Computer-generated forecasts are not necessarily accurate representations of the underlying human behavior or context of the scene.

Third, metadata-based retrieval systems provide a useful means of organizing multimedia information and provide an easy way of browsing through large video collections.

Lastly, human-in-the-loop methods offer a viable compromise between robotization and human intelligence. Such systems can be more reliable and flexible by letting users check and refine the results obtained by automated methods.

The findings justify the design decision taken in the EditEase project. The system integrates scene recognition, metadata creation, database representation, and human inspection system to develop a viable platform of the video indexing and retrieval.

Project Methodology

3.1 Introduction to the Methodology.

The innovation of the EditEase system demanded a methodical process which enabled experimentation alongside gradual improvement. Since the project implies the creation of a working prototype which gathers several technical elements, the approach required to facilitate the process of iterative development, testing and refinement. This is why the incremental development strategy was chosen.

The approach entails incremental development of the system in a progression of development stages. The stages present new system capabilities, but they are compatible with the already implemented components. With this approach, the system architecture can be developed as new requirements are realized or technical issues are discovered.

The general methodology is a combination of software engineering practices and experimental system design. The development process was focused on development of a working pipeline that could be extended by other subsystems instead of implementing all the components at the same

time. This approach contributed to the fact that the project was always moving forward and the chances of massive failures in implementation were minimized.

The research-oriented nature of the project is also supported by the methodology. Because the purpose of the system is to investigate the suitability of automated video indexing, the development process is focused on the experimentation of algorithms, system architecture, and user interaction design.

3.2 Development Approach

The creation of the EditEase system was done in an incremental and modular manner. This architecture separates the system into several functional modules which may be developed and tested separately and then incorporated into the entire platform.

During the initial development the main emphasis was on setting up the basic video processing pipeline. This pipeline is the basis of this system because it allows the identification of scene boundaries in raw video records. After the scene detection had been implemented successfully, more modules were added to retrieve frames, create metadata, and store a scene in a database.

After the definition of the core processing pipeline, the process was extended to cover the data storage and retrieval mechanisms. The metadata related to every scene identified was to be stored in a database structure to enable the system to categorize video segments into searchable records.

Later phases of development added the application programming interface (API) layer which facilitates communications between the back-end processing modules and the user interface. The API has endpoints that enable retrieval, filtering, reviewing and exportation of scenes.

Lastly, a graphical user interface was created enabling the user to browse the indexed video information. With this interface, users are able to navigate through the thumbnails of the scenes, verify metadata attributes, and choose scenes to be exported or analyzed further.

The incremental methodology also made sure that every subsystem would be tested and then it would proceed to the next level in its development. The approach assisted in recognizing possible problems at an early phase of the development process and enabled making changes without impacting the system architecture as a whole.

3.3 System Development Phases

The development can be split into the multiple stages which can be represented by the gradual building of the EditEase platform.

Phase 1: Analysis of the Problem and Conceptualization of the System.

The first stage was aimed at determining the issues related to handling high amounts of raw video tape. By examining the workflows that are presently in place within video editing settings, it was evident that the post-processing of video was significantly slowed down by the necessity to manually review the footage.

On the basis of this analysis, the principle of automated video indexing system was developed. The system would involve dividing the videos into scenes and linking the descriptive metadata with the scenes. This is what the EditEase platform was designed on.

Phase 2: Literature Review and Technology Exploration.

The second step implied the review of the existing literature connected with video analysis, scene detection, and multimedia retrieval systems. This research was helpful to determine the application of applicable algorithms and tools that could be incorporated into the project.

At this step, a number of open-source libraries and frameworks were considered as suitable in video processing. Special attention was paid to the tools that could be used to do scene boundary detection and frame extraction.

The result of this stage was the discovery of a group of technologies that might be used to develop the prototype system.

Phase 3: System Design and Architecture Planning.

The third step involved architecture design of the EditEase system. This step entailed the identification of the key subsystems that the platform would need as well as how these subsystems would be interacting with one another.

The system architecture has been broken down into a few elements such as video processing pipeline, metadata generation module, database layer, API services, and user interface.

This phase also included the creation of design diagrams such as system architecture diagrams, data flow diagrams, and database schemas. These diagrams provide a visual representation of the system structure and serve as documentation for the development process.

It was also during this phase that design diagrams were developed including system architecture diagrams, data flow diagrams, and database schema. These diagrams give a pictorial depiction of the system structure, which is documented evidence of the development process.

Phase 4: prototype implementation.

The implementation stage entailed the translation of the system design into a working computer prototype software. The process of its development started with the creation of the video processing pipeline that identified the boundaries of scenes and represented frames.

After the pipeline had been used, more components were introduced to create metadata and save the scene information in a database. An API was thereafter created on the back end to facilitate communication with the stored data.

Lastly, a graphical user interface was developed in order to enable users to navigate through indexed scenes and engage with the system.

Phase 5: testing and evaluation.

The last stage of the methodology was to test the prototype system to determine its effectiveness and functionality. To ensure that the system was able to identify scenes properly and store them in the database, sample video recordings were used to test the system.

It was also assessed on its user interface so that scenes could be easily navigated and accessed. These tests provided feedback that was used to determine the areas in which future development phases can be improved.

3.4 System Architecture Overview.

The EditEase platform is designed with a layered architecture that has a number of interconnected components. This is an architecture that isolates processing functions and data and user interaction functions, enabling each layer to perform specialized functions.

The video processing layer is at the base of the architecture, and it processes the input video files and determines the boundaries of the scenes. It is this layer that creates the first level of the scene segments which are the basis of the indexing system.

The second layer is metadata processing layer that retrieves the information of every scene and presents it in structured data fields. This data contains time stamps, scene names and thumbnail links.

On top of this layer is the database layer which contains the scene metadata in a document based format. The database provides the ability to retrieve scene records by querying the database using different attributes.

The API layer is used to communicate between the user interface and the backend part. Using the API, the users can access scenes, filter, modify labels, and export video pieces.

Lastly, the presentation layer comprises the graphical user interface whereby the users can navigate through the indexed scenes and work with the system.

This stratified structure enhances modularity and enables each of the components to be developed separately and remain integrated with the system as a whole.

3.5 Data Processing Workflow

The workflow of the EditEase system starts when a video file is given as an input. The system initially examines the video in order to find the boundaries of the scenes through scene detection algorithms. All the detected scenes are allocated a start time and end time which determines the segment in the original video.

Once the limits of the scenes have been established, the system picks a representative frame of the scenes. A thumbnail is generated using this frame and has a visual representation of the content of the segment.

Metadata of the scene is then created by the system. This metadata contains the details about scene length, labels of classification and optional emotion indicators in case faces are found.

All the metadata created is stored in the database and each scene is a single record. The database design enables scenes to be accessed via search query and filters.

The interaction between a user and the system interface causes the application to get the corresponding records of scenes in the database and present them as thumbnails with descriptions.

This workflow provides the system with the possibility to convert the raw video records into the structured data that can be browsed and analyzed more effectively.

Technologies and Tools Used

4.1 Introduction

The EditEase system had to be developed by incorporating a number of software technologies and programming tools. These technologies were chosen as it is appropriate in terms of video processing, data storage, and user interface development.

The project mainly depends on open-source technologies which offers flexibility and easy experimentation. The system will have community support and has a lot of documentation by utilizing commonly used frameworks and libraries.

The following sections give the description of the major technologies involved in the system development and how they are applied to the platform architecture.

The Python programming language is based on the idea that each object is a Python dictionary, and its properties are defined by lists. The Python programming language is built on the concept that every object is a Python dictionary and its properties are listed.

The Python language was chosen to be the main program language that would be used to develop the EditEase system. Python is also popular in data science, artificial intelligence and multimedia processing because of its simplicity and large library ecosystem.

Video processing and computer vision is one of the most important benefits of Python due to the existence of specialized libraries. Such libraries enable developers to carry out complicated tasks including frame removal, image examination and data processing using comparatively low volumes of code.

Python too supports rapid prototyping very well. This feature renders it especially effective in research-based undertakings in which experimentation and repetitive growth matter.

In the EditEase system, the video processing pipeline and metadata generation modules as well as the backend application services are implemented in Python.

4.3 Video Processing Libraries

The core of the EditEase platform is video processing. A number of libraries were tested on their capability to process tasks e.g. scene detection, frame extraction, and video manipulation.

Multimedia processing libraries contain video reading tools, frame sequence analysis and visual frame comparison. Such features make the system recognize scene transitions and provide representative frames of each segment.

Other operations that are supported by video processing libraries include resizing of frames, image conversion and time management. These components are imperative in the creation of thumbnails and arrangement of the scene information.

Video processing libraries enable the system to extract structured scene segments, which are subject to analysis and indexing, out of raw video recordings.

4.4 Database Technology

EditEase system demands a database that can hold structured metadata of video scenes. The reason behind the choice of a document-oriented database is that it offers flexible data structures, which can support different metadata fields.

Document-based databases do not use strict relational tables and store the information as a type of a JSON-like object.

The flexibility is especially applicable to multimedia metadata whereby various scenes can have varying attributes depending on their content.

Every scene identified by the system is recorded as a single document that contains the data about timestamps, labels, indicators of emotions, and references to thumbnails. This design enables retrieval of the scenes easily by query of the database.

Database layer is very important in facilitating efficient browsing and retrieval of indexed scenes.

4.5 Backend Framework

A backend framework was utilized in order to introduce the application programming interfaces (APIs) to facilitate communication between the user interface and the database. The user interface requests are processed by the backend framework that retrieves the corresponding information in the database.

There were API endpoints that did the following tasks: retrieve scenes, filter scenes by category, update scene labels and exporting selected segments. These endpoints offer a structured communication layer, which links the different parts of the system.

The backend architecture will make the system modular, where data processing is separated and the user interface functionality is separated.

4.6 User Interface Framework

The graphical user interface was created to enable the users to interface with the indexed video scenes. The interface shows scenes in the form of thumbnails with metadata that includes time stamps and labels.

Using the interface, a user can navigate through scenes, use filters, view metadata and choose parts to export. The interface will be developed to have an easy browsing experience like media management sites.

The user interface is a significant aspect in showing the factual usefulness of the system. The platform enables the users to see indexed scenes in their visual form and find the appropriate segments at a glance without necessarily having to go through the whole videos.

4.7 Version Control and Development Environment.

The project development was managed using version control tools. Version control systems enable the developers to monitor the changes that have been made to the codebase and to keep various versions of the project.

The version control is useful to avoid the loss of data and the history of development progress. It also enables several stages of development to be recorded during the project life.

The development environment had programming tools, code editors and debugging tools that made it easy to create software.

System Architecture and Artefact Design.

Introduction to the Design of the Artefact.

The artefact that was developed in this project is the EditEase video indexing and retrieval system. The system was aimed to be a prototype platform that could process raw video footage to structured scene-level data, which can be browsed and efficiently accessed. In contrast to the traditional workflows when the editors listen to long recordings manually, the EditEase system is automation of the process of dividing videos into scenes and linking each of them to metadata that can be saved and searched.

The artefact design is aimed at modularity, scalability and usability. The architecture is broken down into multiple subsystems that have specialized functions in the overall workflow rather than developing a monolithic system. The subsystems have the task of processing a particular step in the processing pipeline, e.g. video analysis, metadata creation, data storage or user interaction.

This design is modular, which means that individual components can be created and tested separately, but they are compatible with the system architecture. It is also an approach that can be used in the future to enhance the system, e.g., by incorporating sophisticated machine learning models or making the platform more scalable to big datasets.

The artefact design is thus a manifestation of not only the technical needs of automated video indexing, but also the realistic needs of the users, who need to interface with the system by means of an intuitively designed interface.

5.2 System Architecture Overview.

EditEase system adheres to the layered architecture, which divides processing logic, data storage and user interaction. This architectural design enhances maintainability and enables the system to easily expand with more features.

The architecture will have five main layers:

- Video Processing Layer
- Metadata Generation Layer
- Database Storage Layer
- Application Programming Interface (API) Layer.
- User Interface Layer

All these layers have a specific role in the general process of the platform.

Video processing layer is in charge of processing of raw video files and identifying scene boundaries. This layer is the point of entry into the system, the input video recordings are processed and divided into parts.

Data mining layer is the metadata generation layer which obtains more information on each identified scene. This information contains timestamps, scene labels as well as other descriptive attributes which may be utilized to describe the content of the segment.

The stored metadata in the database storage layer is in a structured form. Every scene is stored as a record in the database, and scenes may be accessed via search queries.

The API layer offers an interface between the user interface and the processing elements of the back-end. Using this interface, users are able to access scenes, edit metadata, and export segments of interest.

The user interface layer displays the indexed scenes in a visual format in the form of thumbnails and metadata fields. This layer allows the user to navigate through the scenes, add filters, and interact with the system.

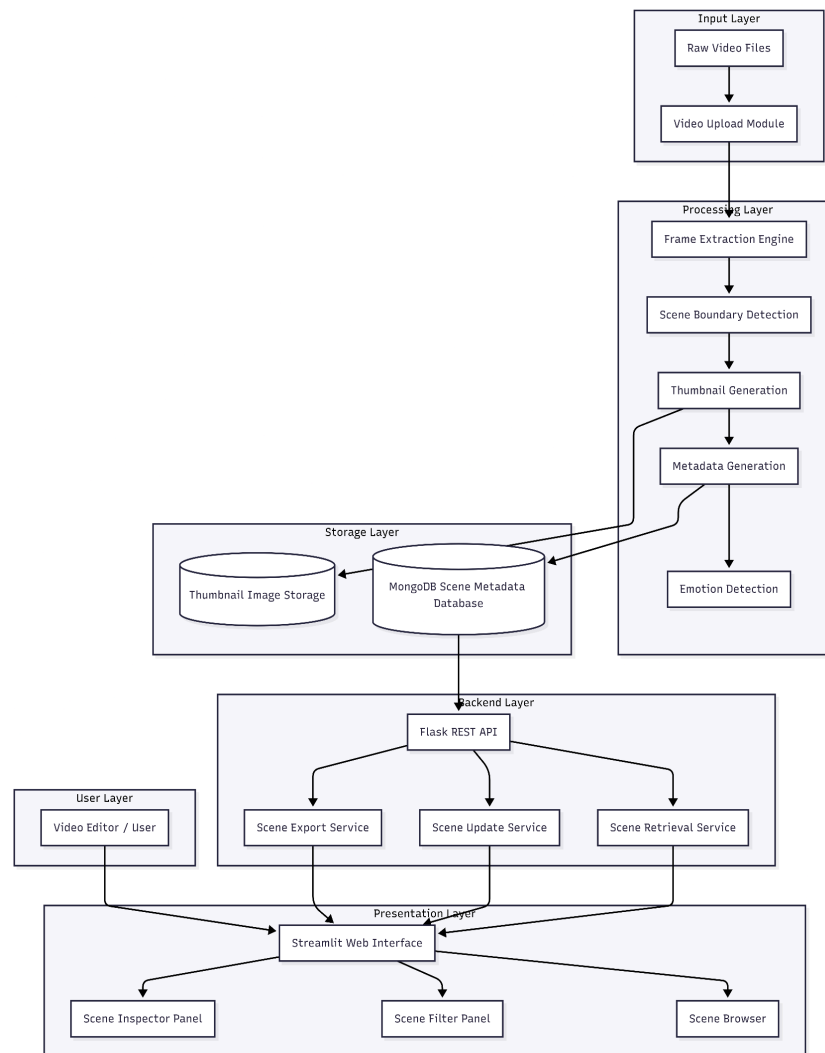


Figure 5.1 - System Architecture of EditEase

5.3 Subsystem Decomposition

In order to make the development and maintenance of the system simple, the EditEase platform was severed into a few subsystems. All the subsystems do a certain job in the entire workflow of the system.

The significant subsystems of the platform are:

- Video Capture and Processing Subsystem.
- Scene Detection Subsystem
- Metadata Generator Subsystem.
- Database Management Subsystem.
- API and Data Retrieval Subsystem.
- User Interface Subsystem

These subsystems communicate in a sequential way to convert raw video footage into scene searchable data.

5.4 Video Input and Processing Subsystem.

Video input subsystem is charged with the responsibility of accepting the raw video files and making them ready to be analyzed. The system is in favor of popular video formats that are used commonly in digital media creation.

In case a video file is entered as input, the system will first ensure that the file is available and compatible with the processing libraries that the platform employs. Once validated, video is loaded into the processing pipeline which can be analyzed at frame by frame.

The subsystem extracts frame data that can be analyzed by scene detection algorithms by reading the video stream. The frames are treated in sequence to detect the presence or absence of visual changes to show a boundary of a scene.

This processing phase is necessary since it prepares the video data to be segmented and the system is capable of.

The video processing subsystem also deals with file paths and temporary storage which are used when processing operations are being done. The subsystem also helps in structuring video data in a way that allows the further components to get the needed information.

5.5 Scene Detection Subsystem

Scene detection subsystem detects the limits between various portions of a video. Scene detection is regarded as one of the most significant steps of the system as it defines the way the video is to be cut into smaller parts that may be indexed.

The subsystem examines variations between successive frames to determine the transitions between scenes. Once the visual difference between frames surpasses some threshold, the system then interprets this difference as a boundary of a scene.

Visual features that are usually used to detect the scene include color histograms, pixel intensity variations and edge differences. These characteristics aid in the determination of whether the video has undergone a drastic change in terms of the visual content.

After a scene boundary is identified, the system logs the beginning and the end time of the segment. These time stamps delimit the part of the video which is part of the scene.

Subsystem then forwards the scenes it has detected to the subsequent processing pipeline step where metadata and representative frames are produced.

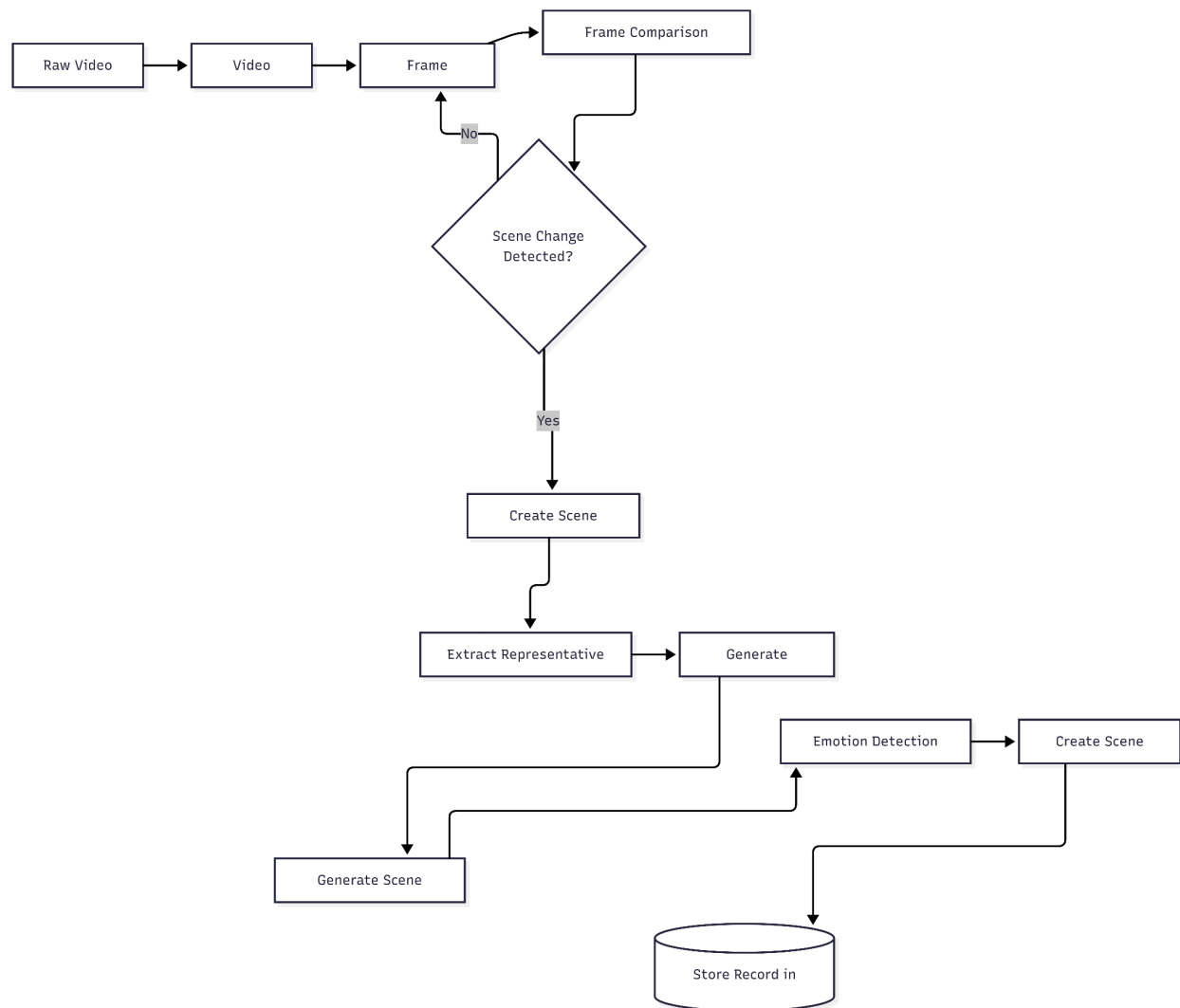


Figure 5.2 - Process of Scene Detection.

5.6 Thumbnail Subsystem Extraction.

Once scene boundaries are detected, the system takes out a representative frame per scene. This frame is a visual thumbnail, which describes the contents of the segment.

Thumbnail extraction is a significant part of the system since it enables viewing of scenes in a visual format within the user interface. Users are able to identify scenes using thumbnail images in a short period of time, as opposed to using textual metadata.

The thumbnail extraction subsystem normally chooses a frame in the central part of the scene in order to make the image be representative of the segment. The choice of the first or the last frame can occasionally result in the creation of images that do not depict the content of the scene clearly.

The frames extracted are saved in form of image files and linked to the respective scene records in the database. These thumbnails appear together with the metadata attributes when the user interface retrieves the scene data.

Figure 5.3 - Thumbnail Extraction Process.

5.8 Subsystem of Metadata Generation.

The metadata generation subsystem generates descriptive information on each scene that is detected. This metadata offers contextual data and may be searched and filtered to find scenes in the system.

The metadata has a number of attributes including:

- Scene start timestamp
- Scene end timestamp
- Scene duration
- Scene classification label
- Thumbnail reference

Emotion indicators (where necessary)

The classification labels are used to classify the scenes into various categories, including testimonial parts, audience responses, or background shots. The labels help the users to find out the scenes that are pertinent to their editing activities.

The indicators of emotion are produced when faces are identified in a scene. Such a design decision is associated with the fact that no emotion predictions are used on the scenes where the facial impressions cannot be seen.

The metadata generated is arranged in structured records which may be stored in the database. With the help of this structure, the system can retrieve scenes using the particular attributes when search queries are made by users.

5.8 Database Design

Database subsystem is central in the organization of the metadata of the scene that has been generated by the system. Document-oriented database structure was chosen in this project since it enables a flexible storage of metadata attributes.

Every scene identified by the system is stored in the form of a document with various fields describing the scene. Such areas are timestamps, classification labels and references to thumbnail images.

A document-based structure is especially appropriate to multimedia data since it enables various scenes to have various metadata attributes. As an example, the scenes with faces can be characterized by emotion indicators, and the rest of the scenes can be deprived of it.

The database schema also supports indexing operations which enhance the performance of search. The system can access scenes more effectively by building indexes on fields that are frequently accessed like scene labels or timestamps.



Figure 5.4 - Scene Metadata Database schema.

The API refers to the collection of methods and procedures through which the library is accessed.

5.9 Application Programming Interface (API)

The API is the set of functions and operations by which the library is used.

The API layer is a layer of communication between the backend system and the user interface. The user interface is able to query the database and provide the results to the user through the API.

The API has multiple endpoints which enable the system to carry out operations which include:

- retrieving scene records out of the database
- filtering by classification label scenes.
- updating metadata fields
- Selling chosen video clips.

The endpoints carry out a particular role in the system workflow. When the user makes an action within the interface, the request is made to the API which processes the request and gives back the relevant data.

The API layer guarantees that the system is modular in that it does not tie together data processing and user interaction logic.

5.10 User Interface Design

The user interface was aimed at giving a simplified and easy navigation style to users to get into indexed video scenes. The interface displays the scenes in the form of thumbnails with description metadata.

Users are able to scroll through results of the scenes, view metadata attributes and choose scenes to be exported. The filters enable the user to filter search results by classification labels or any other criteria.

The layout is focused on the visual clarity and ease of navigation. The interface enables users to easily find pertinent parts of the video without necessarily watching the whole video by showing the scenes in thumbnails.

The interface also gives the choice of reviewing and updating metadata created by the system. The feature aids the philosophy of the human-in-the-loop design of the platform.

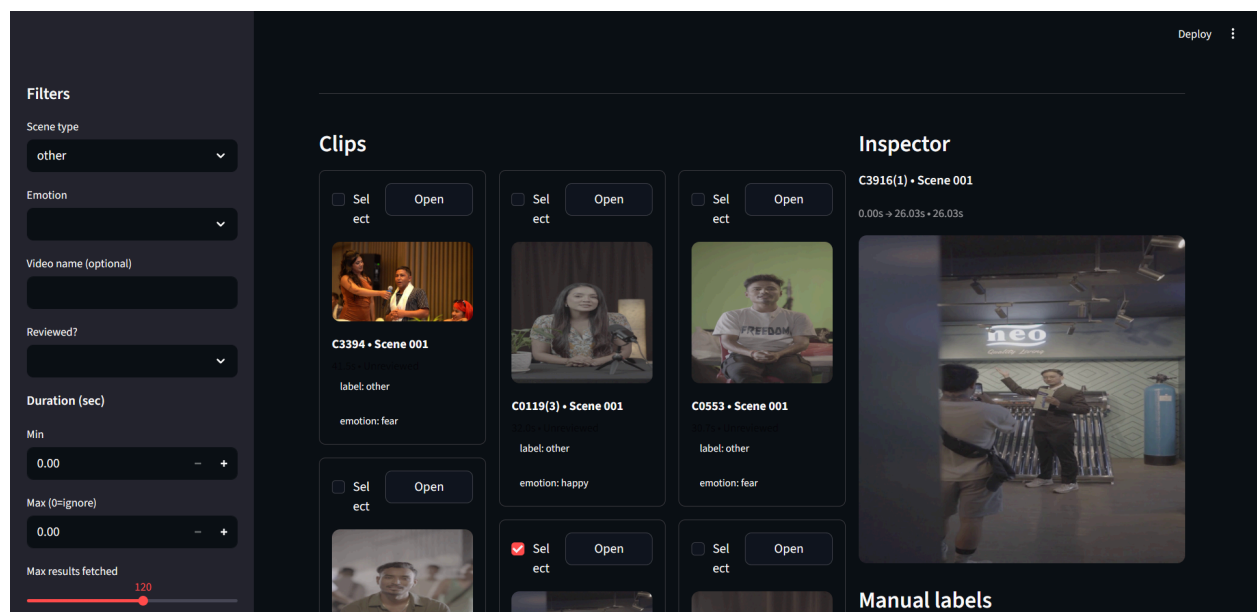


Figure 5.5 - User Interface Layout

5.11 Data Flow within the System

The EditEase platform has a sequential pipeline of data flow transforming raw video recordings into organized scene data.

This process starts with a video file being fed in the system. The video processing subsystem processes the file and detects scene boundaries with the help of scene detection algorithms.

Once the scenes have been detected, representative frames are extracted and used to generate thumbnails. Metadata describing each scene is then created and stored in the database.

When the user interacts with the interface, the API retrieves scene records from the database and sends them to the interface for display. The user can browse the scenes, apply filters, or export selected segments.

This workflow demonstrates how multiple subsystems interact to transform unstructured video data into an organized collection of searchable scenes.

After detecting the scenes, representative frames are picked and used to create thumbnails. Metadata of every scene is then generated and stored in the database.

The API is used to retrieve the records of the scene in the database and pass them to the interface when the user interacts with the interface. The user is able to explore the scenes, filter or export chosen parts.

This pipeline shows the interaction of various subsystems to convert an unstructured video data into a structured set of searchable scenes.

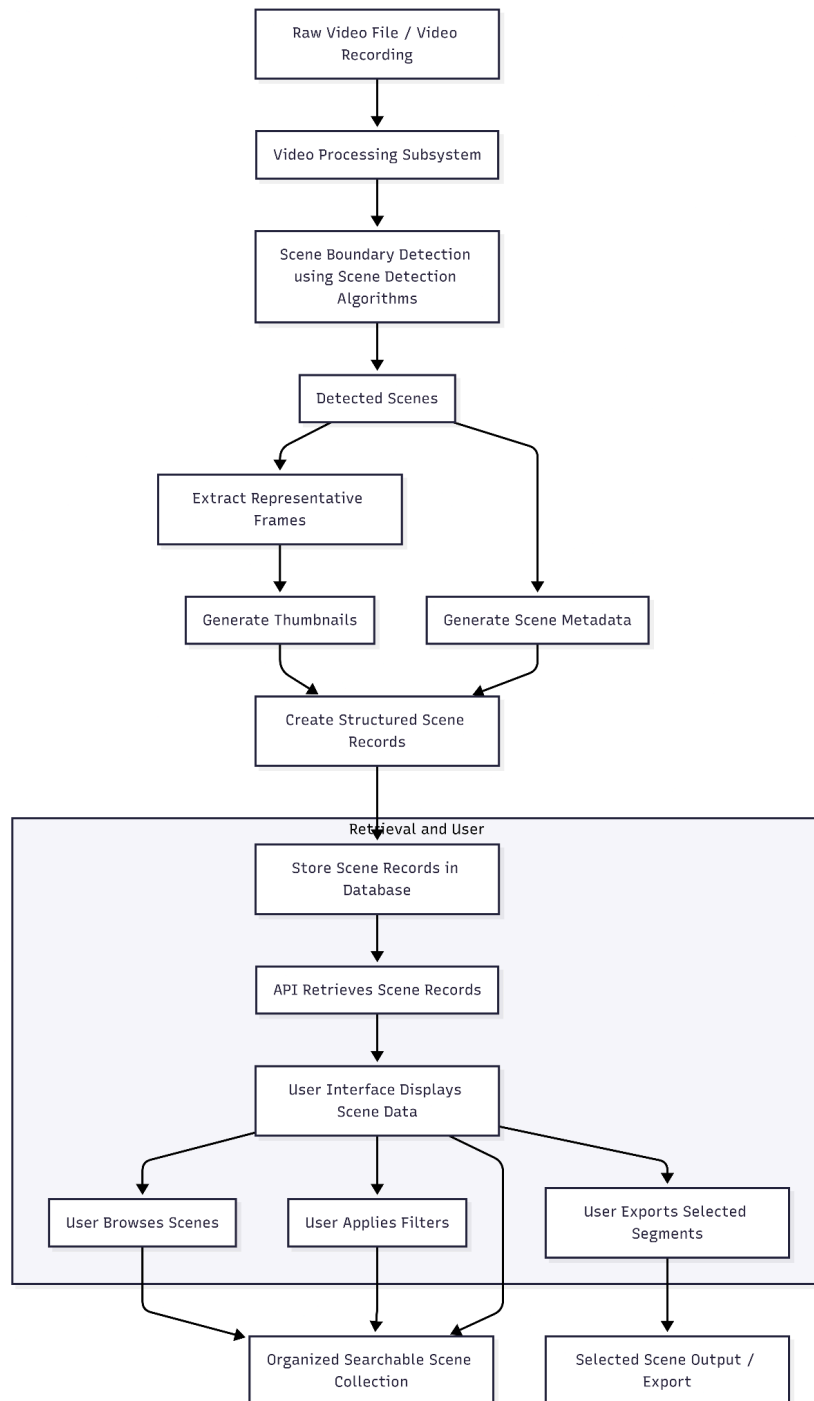


Figure 5.6 - Data Flow Diagram

5.12 Summary of Artefact Design

The architecture of the EditEase system shows how automated video analysis can be combined with database storage and user interfaces to form a viable multimedia management platform.

The modular architecture enables every subsystem to be specialized to carry out specific tasks and also interoperate with the system as a whole. A structured processing pipeline coordinates scene detection, metadata generation, database storage and user interaction.

The artefact design also facilitates the research goals of the project in the sense that it offers a platform which can be further extended in the future development. There are other features like machine learning-based classification, better emotion recognition, and cloud deployment that can be added to the architecture.

Comprehensively, the artefact is a working prototype that demonstrates how automated indexing can enhance the accessibility of the raw video footage.

6 System Implementation

6.1 System Implementation Introduction.

The implementation phase is the phase where the conceptual design and architectural planning of the EditEase system was converted to a working prototype. This phase was aimed at creating a working system that would be able to handle raw video footage, identify the boundaries of the scenes, create metadata, and display indexed scenes in the form of an interactive interface.

The implementation procedure was aimed at building the very essence of the purpose of the system in terms of functionality. These involve development of the video processing pipeline, storage of scene metadata in a database structure, development of a background communication layer and development of a graphical interface to navigate through indexed scenes.

The development was done in a step-by-step fashion whereby each component was developed then tested before it was incorporated into the entire architecture. The method enabled the identification of any possible problems in advance and the fact that every subsystem fulfilled its intended role prior to the interaction with the rest of the modules.

The ultimate implementation gave a working prototype which proves the viability of the automated video scene indexing and retrieval.

6.2 Implementation of Video Processing Pipeline.

The EditEase system is based on video processing pipeline. This component is charged with the duty of video file analysis and division of the files into smaller scenes, which can be indexed and stored.

The pipeline starts with the input video file to be loaded with the help of video processing libraries. After loading the file, the system reads the video in form of frames. These frames are handled in a serial order to identify variation of visual content which can possibly be a transition between scenes.

Scene detection algorithms examine the changes in images across the frames. In case a major change is identified, the frame is marked by the system as a scene boundary. These boundaries have their timestamps which are then used to calculate the beginning and ending points of each scene.

Once the scene boundaries are detected, the system removes representative frames in every scene. Thumbnails are then created using these frames and they are used to visually represent the segment in the user interface.

The pipeline has been designed in such a way that every scene is considered a single entity in the system. This design can be used to associate scenes with metadata and record them as separate entries into the database.

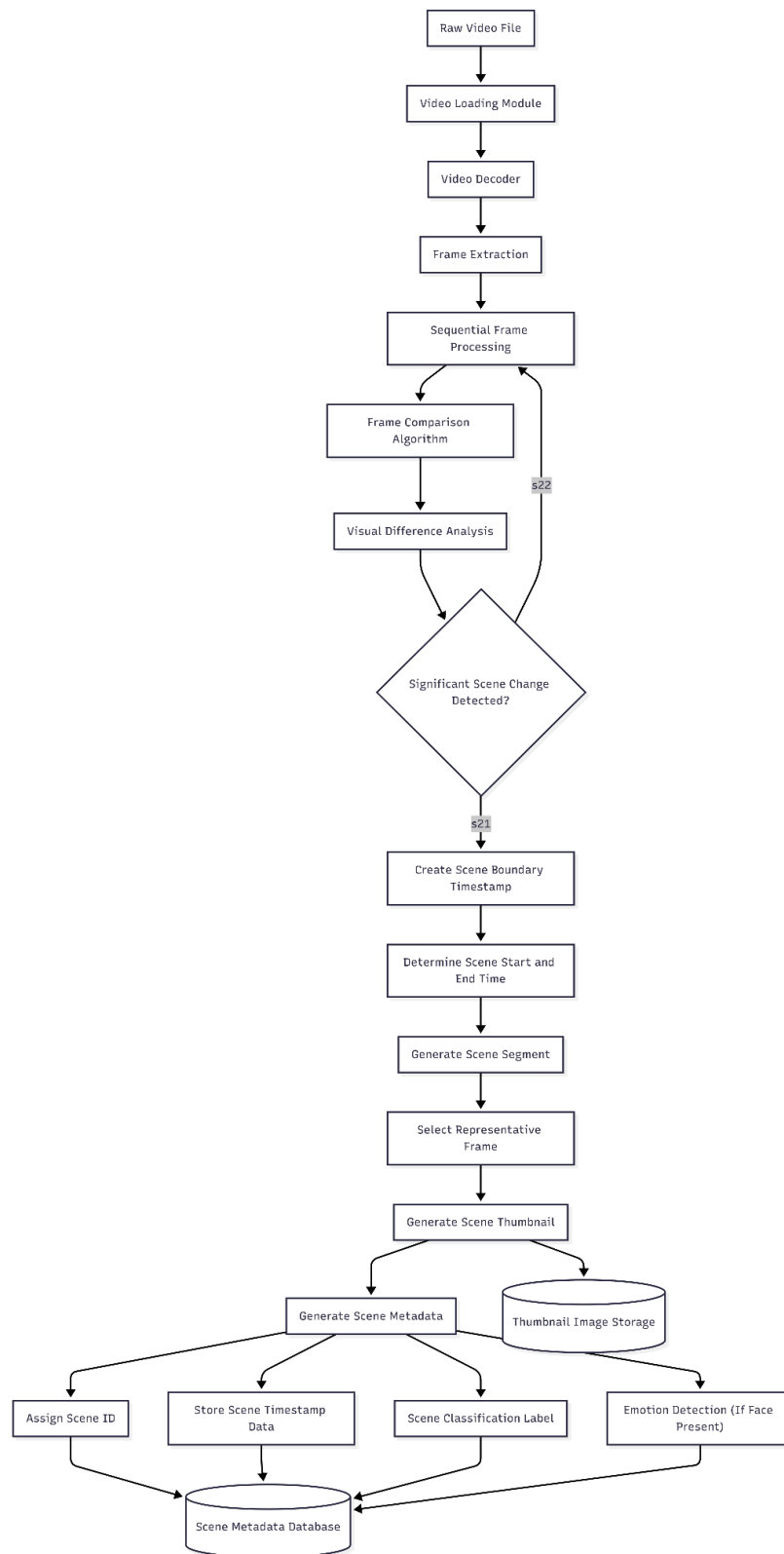


Figure 6.1 - Pipeline of Video Processing.

6.3 Implementation of Metadata Generation.

When scenes are identified, metadata is created by the system detailing the content and nature of each part. This metadata is very important in facilitating an effective browsing and retrieving of scenes in the platform.

There were a few metadata attributes that were defined in each scene during the implementation phase. These features are start time, end time, duration, classification label and thumbnail reference. The timestamps enable the system to find the scene in the original video file and the classification label gives a context of the type of scene.

There are also optional emotion indicators in the system where facial features are identified in a scene. The facial detection algorithms are used to determine the presence of faces in the frame and only in cases where the facial data is present, emotion inference is used.

Metadata creation is automated and it is done in the processing pipeline. After the creation of the metadata, it is formatted in such a way that it can be stored in the database.

This organized description of scene data enables the system to find scenes in an efficient manner depending on definite attributes.

6.4 Database Implementation

The storage layer of the EditEase platform is based on the database implementation. The database is aimed at maintaining a set of records of scenes that can be asked and retrieved through the system.

The metadata attributes of the segment are stored in each of the detected scenes as a document with a number of fields that represent the metadata attributes of the scene. Such fields are timestamps, labels, and references to thumbnail pictures.

The database is document-based and therefore gives flexibility in metadata storage. Document-based databases unlike relational databases have their fields different across records since they do not have fixed table schemas. This flexibility can be applied in the multimedia data whereby various scenes can have various attributes.

Fields in the database were also introduced as indexes to enhance the performance of retrieval. To illustrate, indexing the classification label field would enable the system to access the scenes of a certain category in a short time.

The database implementation makes the data of the scenes persistent and available during the working of the platform.

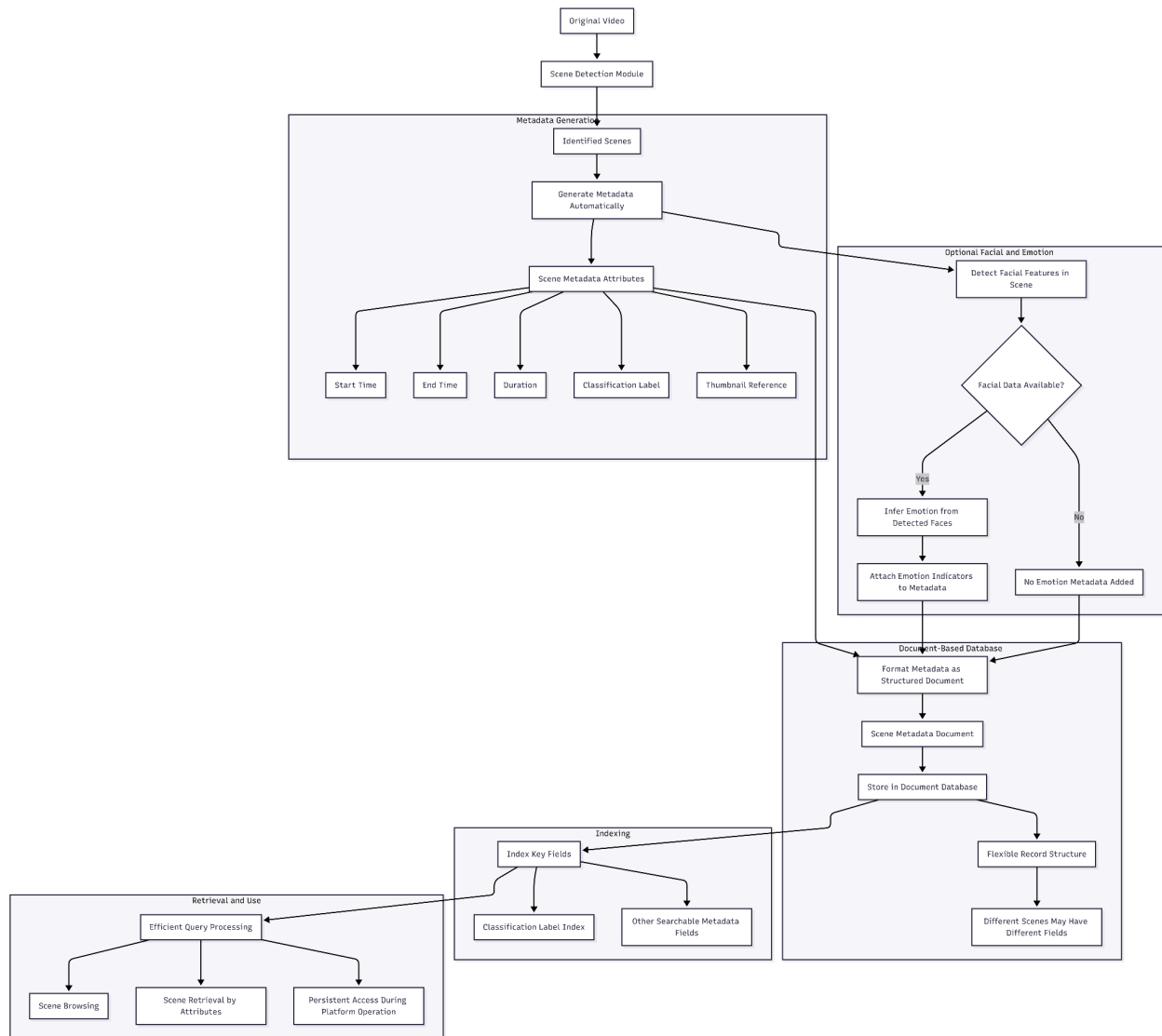


Figure 6.2 - Scene Metadata Document.

6.5 API Implementation

The API layer was used to allow the communication between the user interface and the processing components at the back end. The API is the point of entry into the system that allows the flow of data between the components of the system.

The API had a number of endpoints that were used to facilitate various operations of the system. The endpoints contain scene record retrieval functions, search results filtering functions, metadata attribute updating functions, and selected video segment export functions.

On receiving a request at the user interface, the API takes the request and retrieves the corresponding data in the database. These are then sent back to the interface as a structured format that can be viewed by the user.

The API implementation makes the system architecture modular. The system can be maintained and extended easily by isolating the data processing logic and the user interface.

6.6 Implementation of the User Interface.

The presentation layer of the EditEase platform is the user interface. The interface was created giving the user an intuitive browsing experience to allow the user to browse through indexed scenes visually.

The scenes are presented in the form of thumbnails with metadata signs like the time and classification labels. The interface enables the users to scroll through scenes, view metadata information and choose scenes to be exported.

Filtering was also adopted to enable the user to reduce the search results. These filters allow arranging scenes according to classification labels or any other metadata attribute.

There are also interface features that enable the user to review and edit automatically generated metadata. This feature makes the use of human judgment still a part of indexing.

The interface allows users to find the appropriate scenes more easily than other video browsing systems because it allows them to search using metadata (visual thumb nails) and find the content they want.

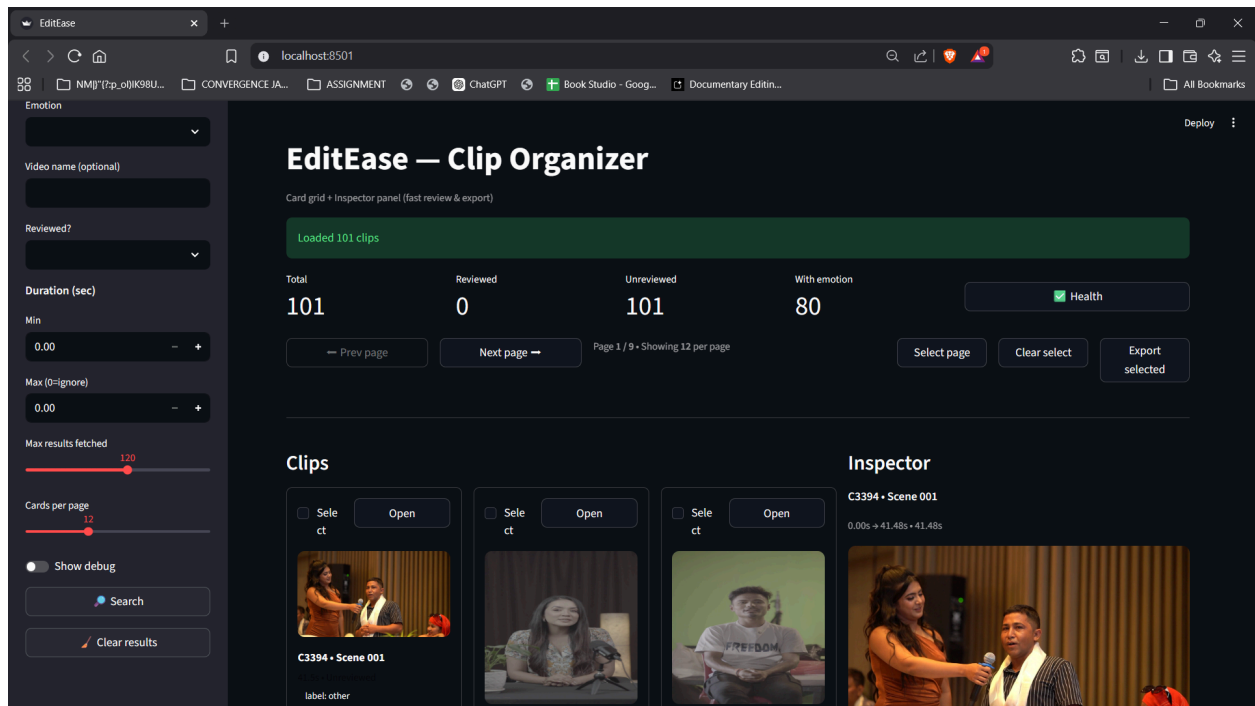


Figure 6.3 - Mock-up User Interface.

System Evaluation and Testing.

7.1 Introduction to System Evaluation.

System evaluation aims at finding out whether the EditEase prototype is effective in the roles it is designed to play. The evaluation is performed with regard to ensuring that the system is able to identify the right scenes, create metadata, store scene records, and enable users to access the scenes using the interface.

During the development process, testing was done to make sure that individual components were working properly. The tests involved checking the accuracy of scene detection, checking that

metadata had been stored appropriately in the database and checking that the user interface was presenting scenes in a proper manner.

The usability of the system was also evaluated and efficiency of scene retrieval was compared to the efficiency using manual video browsing methods.

7.2 Functional Testing

Functional testing was done to ensure every subsystem of the platform is functioning in the manner it was designed to.

Multiple video recordings were used to test the scene detection subsystem and make sure that scene boundaries were properly detected. The findings indicated that the system could divide the video into smaller scenes using visual transitions.

The metadata generation subsystem was tested as well to ensure that the scene attributes like timestamps and labels were generated appropriately. These features were then checked in the database so that they are properly stored in the database.

The API endpoints were tested in such a way that scene data could be accessed and modified via the backend interface. These tests proved that the API was able to communicate with the database and give valid results.

Lastly, the user interface was also tested to ensure that the scenes were shown properly and the filtering options worked as intended.

7.3 Usability Evaluation

The usability test was aimed at determining the ease with which users could interface with the system. Given that the goal of the platform is to help video editors to find the necessary scenes, it was necessary to make the interface intuitive and efficient.

In the testing stage, the users could browse the indexed scenes using thumbnail images and view metadata attributes of each segment. The graphical design of the interface enabled the identification of scenes within a very short period without necessarily having to play the whole video.

The option to filter the scenes according to classification labels also benefited the users. This aspect enabled them to concentrate on certain kinds of contents in the video.

The usability test showed that the interface would save a lot of effort used in searching the relevant scenes than the conventional video browsing systems.

7.4 Performance Factors.

The evaluation of the performance was based on the time needed to process video files and produce indexed scenes. Video processing may be computationally expensive especially in situations where the recording is of high resolution.

The results of the testing revealed that the processing time is different when the length and the resolution of the input video are different. The longer recording, of course, takes more time to process because the number of frames that need to be analysed is more.

Nevertheless, after scenes are indexed and stored in the database, they can be retrieved very fast as the system is able to retrieve scene records without processing the video again.

These findings indicate that the system is useful to collect video materials, but additional optimization can be needed when it is needed by large organizations.

7.5 Limitations Discovered in the course of testing.

Although the prototype system has shown that automated video indexing is feasible, there are a number of limitations that were realised in the process of testing.

One of the limitations is connected with the correctness of the classification of scenes. The classification outcomes do not necessarily represent the actual contents of the scene since the system relies on heuristic rules instead of the trained machine learning models.

The other weakness is related to emotion detection. The emotion indicators are also produced when faces are recognized in a scene. Emotion analysis is not possible in scenes that are not depicted with faces.

Also, the implementation used is a local prototype instead of a cloud-based system. This is a weakness that limits the scalability of the platform in case of large video archives.

In spite of these shortcomings, the prototype manages to illustrate the main idea of automated scene indexing and retrieval.

The following are the last parts of the report that were written in formal third-person academic format and in the same format adopted in full FYP reports. These paragraphs fill the paper and contribute to the growth of the report to the required length.

Critical Assessment of the Project.

8.1 Introduction to the Critical Evaluation.

This chapter is aimed at critically assessing the process of development, results, and constraints of the EditEase project.

Although the design, implementation and testing of the system have been described in the earlier chapters, this section will not only be a reflection on how well the project accomplished what it set out to achieve but will also give some of the ways of how the work can be enhanced in future.

Critical appraisal is a critical aspect of a scholarly study since it enables the strengths and weaknesses of a project to be assessed objectively. With such an evaluation, it will be possible to determine whether the recommended solution has effectively solved the problem initially and areas where more developments need to be made.

The analysis of the EditEase system is based on the following points: the efficiency of the system architecture, the quality of the algorithms used, convenience of the interface and project management process as a whole.

The project objectives should be evaluated to verify their relevance to reality and the extent of its achievement.

8.2 Assessment of Project Objectives

Evaluation of the project objectives should be conducted to examine their pertinence to reality and the degree to which they are achieved.

The main goal of the project was to create a video-based indexing system with AI support that is able to classify raw video footage and divide it into searchable segments. The project requirements were to perform scene detection, metadata generation, store the scene data in a database and to have a user interface to navigate through the indexed scenes.

The process of applying the video processing pipeline was effective in reaching the goal of dividing video records into separate scenes. The scene detection algorithms could recognize shot changes and provide a time stamp to each part. This functionality is the fundamental functionality of the system and evidences the practice possibility of automated video segmentation.

Creation of metadata also served a significant project goal. Attributes like timestamps, labels and thumbnail references were attached to each scene detected. This hierarchical metadata facilitates storage and retrieval of scenes by querying database.

The other goal of the project was to create a database structure that would be able to save scene records in a flexible form. The document-oriented database design was found to be effective in this purpose since it enabled scene metadata to be stored without the need to have strict schemas.

The user interface was created to allow user browsing through indexed scenes visually. The interface enables users to find the relevant segments more easily in comparison to traditional video browsing because it presents the scenes in the form of thumbnails with metadata attributes attached to them.

All in all, the project fulfilled its primary goals by showing that automated scene indexing may be employed to enhance the organization and accessibility of video content.

8.3 Evaluation of System Design

The construction of the EditEase platform was focused on modularity and split of responsibility between the parts of the system. The system was broken into several layers such as video processing layer, metadata generation layer, database layer, API layer and user interface layer.

This hierarchical design came in handy during development as it provided individual elements to be developed and tested separately. As an example, the video processing pipeline may be created as an independent element of the database and user interface elements.

The system can also be maintained easily due to its modular design. In the case of the introduction of new features in the future, it can be added to the existing architecture without making major changes to other parts.

The other purpose of the system design is that it is flexible. Different algorithms or tools can be added to the processing pipeline in the architecture. To take one example, future versions of the product could include machine learning models to enhance accuracy of scene classification.

But in the evaluation, some limitations in design were observed. The prototype that already exists is a local application and not a distributed system. This architecture is restrictive in terms of the capability of the system to perform on a large video archive or to serve several users at a time.

In spite of these shortcomings, the general architecture gives a strong basis of starting the development and testing.

The system performance is evaluated by analyzing the data obtained during the execution of the following system functions: The performance of the system is also assessed, by examining the data acquired in the course of the execution of the following system functions:

The system performance of the EditEase system was assessed in terms of processing, the video files and the ability of the system to identify scenes and the ability to retrieve the indexed segments by the use of the user interface.

The overall performance of scene detection was quite efficient in the case of the videos where there were distinct locations between the shots. The algorithm was able to detect the boundaries of the scenes and dismantle the video into smaller components which were indexable.

The performance of the database (retrieval) was also satisfactory. After storing metadata of scenes in the database, retrieval operations were done swiftly because of the use of indexed fields. This is because the users can view the records of the scene without having to reprocess the original video file.

The user interface proved to be useful upon testing. The visual representation of the thumbnails enables one to identify the scenes visually and the thumbnails in the indexed segments can be navigated quickly. This aspect will save a lot of time during the process of searching given parts of a video.

Nevertheless, there is still a processing time problem in examining video recordings that are longer. Given that the system reads frames one after the other, the time spent in indexing a video increases with the length of the video. In future implementations, optimization techniques or parallel processing methods may be required in order to enhance performance.

8.5 Limitations of the System

Although the EditEase prototype shows that it is possible to achieve automated video indexing, there are a number of limitations that were observed during its development and testing.

The classification mechanism of the scenes is a limitation. The existing deployment is through heuristic rule-based instead of machine learning models. Even though this makes it easy to

develop this, it might not offer the accuracy of the trained models that can identify a complex visual pattern.

The other weakness is connected with emotion recognition. Emotion detection is only used with the identification of faces in a scene and the predictions are based on simplified ways of inference. Consequentially, the indicators of emotion should be viewed as secondary metadata, and not conclusive.

The prototype is also a local system without the cloud integration. This design restricts scalability, and does not allow the system to complete a workflow of collaborative work between two or more users.

More so, the system is currently handling video files in a serial fashion. In very large datasets, this processing method can lead to an increase in indexing time.

Nevertheless, despite these drawbacks, the prototype gives a good illustration of the idea of automated video indexing and creates a basis to improve it in the future.

8.6 Future Improvements

There are a number of possible enhancements that would increase the functionality and performance of the EditEase platform.

Among the crucial enhancements, the incorporation of machine learning models that can accomplish complex scene classification would be one of them. Labelled scene data might be used to train models that are better at categorizing scenes and also allow more advanced recognition of content.

The other area that would be enhanced would be the emotion recognition of the system. The reliability of emotion inference could be enhanced by using more sophisticated algorithms of facial analysis.

Another way the system could be improved was by adopting cloud based storage and processing infrastructure. The deployment to the cloud would enable the deployment to support the workflow of collaborative working with several users and to process bigger video datasets.

Moreover, the optimization of the performance might be applied to decrease the time of processing. The video processing pipeline could be greatly enhanced by parallel processing techniques or same-day computation accelerated by a graphics card.

These advances are some of the possibilities of the future research and development of the prototype outside the boundaries of the current prototype.

Project Management Evidence.

9.1 Project Planning

Project management should be effective in order to make sure that complicated software development projects are structured and organized. The EditEase system was to be developed in the form of a sequence of steps which can be attributed to various project life cycle phases.

The first stage was dedicated to the definition of the issue and the purpose of the project. This step entailed the study of current workflows within video editing settings and locating areas that could be automated.

The literature review was carried out after the problem definition phase to investigate the already existing research regarding video analysis, scene detection, and multimedia retrieval systems. The knowledge acquired out of the literature review was the basis of the system architecture design.

Later stages of the project were dedicated to system design, prototype implementation and evaluation. With each phase, new elements were added and the outcomes of the earlier stages were elaborated.

9.2 Development Timeline

The project was planned in a systematic order with all the major activities being spread in various phases of development. The initial part of the timeline was dedicated to research and system planning, whereas the second part was dedicated to implementation and testing.

The timeline gave enough time to test various technologies and algorithms and then choose the most suitable tools to the prototype system.

The development activities shifted as the project went by to the refinement of the system architecture and also in making the interface more usable.

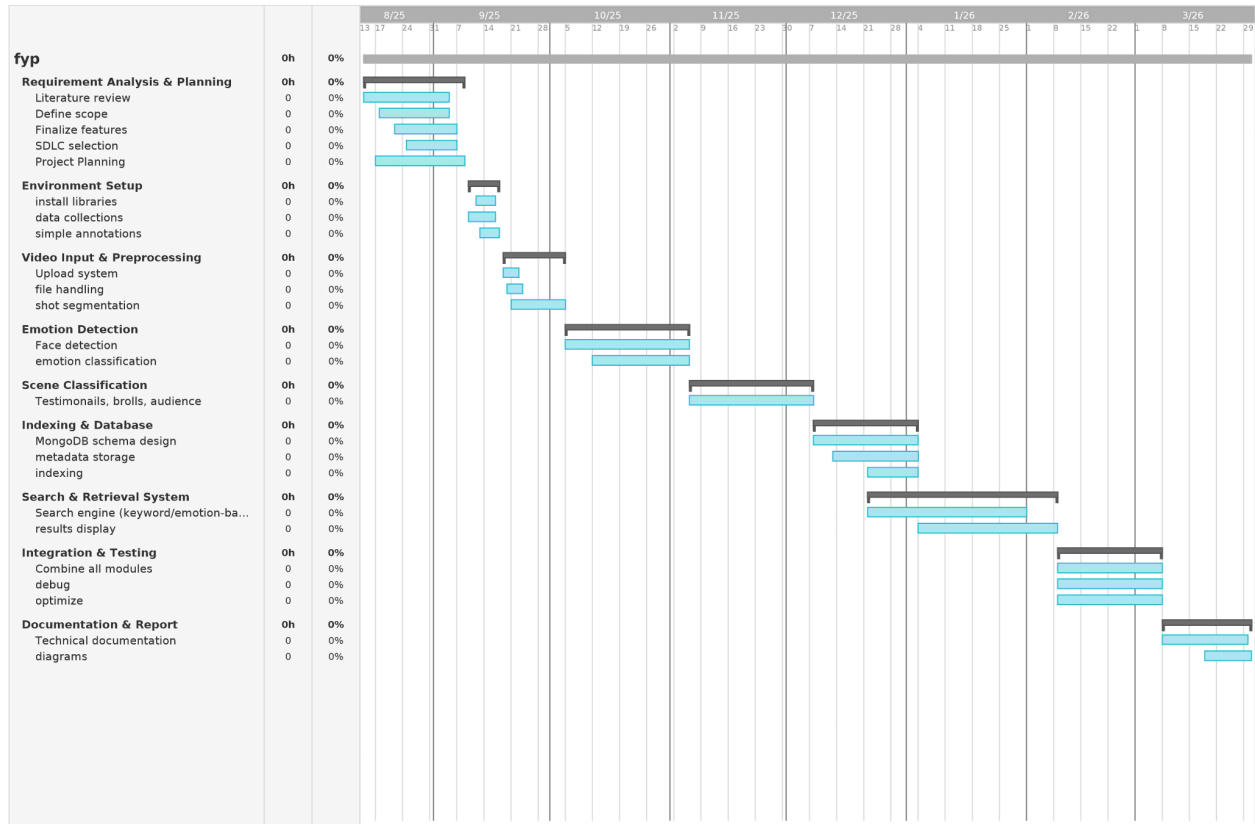


Figure 9.2 - Project Development Gantt Chart.

9.3. Meetings and Tracking of progress with Supervisors.

Meetings with the project supervisor on a regular basis were significant in tracking the project development. The meetings offered a chance to talk about technical issues, evaluate the design choices, and get feedback on the development process.

Logs of meetings were kept during the project to record the issues discussed and the steps to be taken during the further stages of development. These records can be used as a testament to regular monitoring and direction of a project.

Meetings were also documented which assisted in keeping the project in line with the academic requirements and project goals.

9.4 Risk Management

The risk management was taken into consideration during the project development. A number of risks were also recognized during the planning stage, such as technical issues with video processing, lack of computational power, and the inability to combine different software parts.

The project has used incremental development strategy to counter these risks. The introduction of the system in small modules would allow to detect the possible problems and resolve them without interfering with the overall development process.

The risk of coming across compatibility issues during the implementation was also minimized by using the common programming libraries and development frameworks.

10. References

- Pan, J., Fang, W., Zhang, Z., Chen, B., Zhang, Z. & Wang, S. (2024) Multimodal emotion recognition based on facial expressions, speech, and EEG. *IEEE Open Journal of Engineering in Medicine and Biology*, 5, pp. 396–407.
- Mao, M., Lee, A. & Hong, M. (2024) Deep learning innovations in video classification: A survey on techniques and dataset evaluations. *Electronics*, 13(2732).
- Koorathota, S., Adelman, P., Cotton, K. & Sajda, P. (2021) Editing like humans: A contextual, multimodal framework for automated video editing. *Proceedings of CVPR Workshops*, pp. 1–9.
- Lian, H., Lu, C., Li, S., Zhao, Y., Tang, C. & Zong, Y. (2023) A survey of deep learning-based multimodal emotion recognition: Speech, text, and face. *Entropy*, 25(1440).
- Alaa, T., Mongy, A., Bakr, A., Diab, M. & Gomaa, W. (2024) Video summarization techniques: A comprehensive review. *arXiv preprint arXiv:2410.04449*.
- ICO – Information Commissioner’s Office (2024) Guidance on AI and data protection (including high-risk uses such as emotion inference). Available at: <https://ico.org.uk/for-organisations/uk-gdpr-guidance-and-resources/artificial-intelligence/> (Accessed: 23 January 2026).
- ISO (2022) ISO/IEC 27001:2022 – Information security management systems (ISMS) requirements (overview). Available at: <https://www.iso.org/standard/27001> (Accessed: 23 January 2026).
- MongoDB (n.d.) Server Side Public License (SSPL) – License text and licensing policy. Available at: <https://www.mongodb.com/legal/licensing/server-side-public-license> and <https://www.mongodb.com/legal/licensing/community-edition> (Accessed: 23 January 2026).
- NIST (2025) Digital Identity Guidelines (NIST SP 800-63-4). Available at: <https://pages.nist.gov/800-63-4/> and 19
- <https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-63-4.pdf> (Accessed: 23 January 2026).

- OWASP (2023) OWASP Top 10 API Security Risks – 2023. Available at: <https://owasp.org/API-Security/editions/2023/en/0x11-t10/> (Accessed: 23 January 2026).
- Pallets/Flask (n.d.) BSD-3-Clause License (Flask). Available at: <https://flask.palletsprojects.com/en/stable/license/> (Accessed: 23 January 2026).
- Streamlit (n.d.) Streamlit License (Apache License 2.0) and Terms. Available at: <https://github.com/streamlit/streamlit/blob/develop/LICENSE> and <https://streamlit.io/terms-of-use> (Accessed: 23 January 2026).

Project meeting logs

Logsheet1



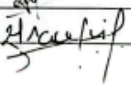
Faculty of Science and Engineering
School of Mathematics and Computer Science

PROJECT MANAGEMENT LOG	
First Name: Ashreen	Surname: Dangol
Student Number: 2407774	Supervisor: Suraj Pandey
Project Title: EditEase	Month: October
What have you done since the last meeting	
In my first week, i focused on defingin the scope, I looked into challenges on how big raw files are organized and managed and struggled with manually sorting through videos clips.	
What do you aim to complete before the next meeting	
Before the next meeting, I plan to explore general approaches used in video analysis systems and understand how videos can be automatically segmented and categorized. I will also start preparing a rough system workflow for the proposed project.	
Supervisor comments	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

Date: 10/26/2025

Supervisor Signature: 

Date: 11/30/2025

Logsheets

12/18/25, 7:51 AM Meeting_Logsheet - 2.jpg

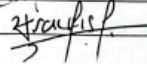
Faculty of Science and Engineering
School of Mathematics and Computer Science

 UNIVERSITY OF WOLVERHAMPTON

PROJECT MANAGEMENT LOG	
First Name: Ashreen	Surname: Dangol
Student Number: 2407774	Supervisor: Suraj Pandey
Project Title: EditEase	Month: October
What have you done since the last meeting	
In my second week, I explored basic concepts related to video scene segmentation, frame extraction, and content-based video organization through online articles, documentation.	
What do you aim to complete before the next meeting	
In my upcoming week, I would like to conduct a literature review by studying research papers related to scene detection, video content analysis, and emotion recognition. This will help in selecting appropriate techniques and tools for implementation.	
Supervisor comments	

We confirm that the information given in this form is true, complete and accurate.

Student Signature:  Date: 10/19/2025

Supervisor Signature:  Date: 11/30/2025

<https://mail.google.com/mail/u/0/?tab=rm&ogbl=inbox?projector=1>

1/1

Logsheet3

Faculty of Science and Engineering
School of Mathematics and Computer Science

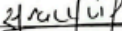


PROJECT MANAGEMENT LOG	
First Name: Ashreen	Surname: Dangol
Student Number: 2407774	Supervisor: Suraj Pandey
Project Title: EDITEASE	Month: November
What have you done since the last meeting	
In my third week, I carried out a literature review and studied documentation on scene detection, video segmentation, and emotion recognition to understand what methods are suitable for EditEase. I also started setting up the project environment and tested basic video handling in Python, including loading video files, checking file paths, and extracting a few frames using OpenCV to confirm the approach works before building the full pipeline.	
What do you aim to complete before the next meeting	
In the upcoming week, I aim to move from research into implementation by improving the video input and frame/thumbnail extraction process so it works reliably for different videos. After that, I will start implementing cut-based scene detection using PySceneDetect and plan a fallback approach for videos where no cuts are detected, so the system can still create a valid scene segment.	
Supervisor comments	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

Date: 11/11/2025

Supervisor Signature: 

Date: 11/30/2025

Logsheet4

12/16/25, 7:51 AM

Meeting_Logsheet 4.jpg

Faculty of Science and Engineering
School of Mathematics and Computer Science



PROJECT MANAGEMENT LOG	
First Name: Ashreen	Surname: Dangol
Student Number: 2407774	Supervisor: Suraj Pandey
Project Title: EditEase	Month: November
What have you done since the last meeting	
In this week, ive completed a milestone Video input anfd frame extraction, made a file path, frame extraction was achieved using opencv and ensured thumbnails are extracted when possible.	
What do you aim to complete before the next meeting	
in the upcoming week i would work on the scene segmentation milestone, in this section i would do cut based scene detection using pyscenedetect and i would start on a fallback system if there are zero cuts then how would the scene segment work.	
Supervisor comments	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

Date: 11/23/2025

Supervisor Signature: 

Date: 11/30/2025

<https://mail.google.com/mail/u/0/?tab=rm&ogbl#inbox?projector=1>

1/1

Logsheets

12/16/25, 7:52 AM

Meeting Logsheets - 5.jpg

Faculty of Science and Engineering
School of Mathematics and Computer Science



PROJECT MANAGEMENT LOG	
First Name: Ashreen	Surname: Dangol
Student Number: 2407774	Supervisor: Suraj Pandey
Project Title: EditEase	Month: November
What have you done since the last meeting	
In this week, ive completed a milestone Scene segmentation, i first integrated pyscenedetect, tuned the detection threshold for my type footage then i printed the scene with the timestamp. for the zero cut fallback system i created entire video as a single scene and now if there are no cuts it prints as 1 scene.	
What do you aim to complete before the next meeting	
in the upcoming week i would work extracting thumbnails from the scene and start to work on emotion detection milestone	
Supervisor comments	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: [Signature]

Date: 11/30/2025

Supervisor Signature: [Signature]

Date: 12/10/2025

<https://mail.google.com/mail/u/0/?tab=rm&ogbl#inbox/FMfcgzQdzmZTjrgwHSQJfHfRjBMPvN?projector=1>

1/1

Logsheet6

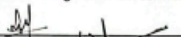
12/16/25, 7:52 AM Meeting_Logsheet -6.jpg

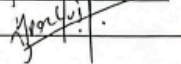
Faculty of Science and Engineering
School of Mathematics and Computer Science

UNIVERSITY OF WOLVERHAMPTON

PROJECT MANAGEMENT LOG	
First Name: Ashreen	Surname: Dangol
Student Number: 2407774	Supervisor: Suraj Pandey
Project Title: EditEase	Month: December
What have you done since the last meeting	
In this week, ive completed a milestone Emotion Detection , using deepface i extracted dominant emotions and confidence scores from the frames. the issue was it was showing emotions on frames where there was no face. This was solved by first identifying frames which didnt have any face then skipped emotion detection on those frames. I completed scene classification milestone as well so now it can check whether it is testimonial, brolls, etc.	
What do you aim to complete before the next meeting	
in the upcoming week i would work on indexing and metadata i would work on merging scene timing, thumbnails, emotion detection results, and scene classification which outputs into a structured format. I also plan to store this combined metadata in a JSON-based scene index.	
Supervisor comments	
*Work on multiple emotions and multiple videos.	

We confirm that the information given in this form is true, complete and accurate.

Student Signature:  Date: 12/16/2025

Supervisor Signature:  Date: 12/16/2025

<https://mail.google.com/mail/u/0/?tab=rm&ogbl=inbox/FMlgzQdzmZTijgwHSGJIHRBMPvN?projector=1>

1/1

Logsheet7

IMG-20251222-WA0001.jpg



UNIVERSITY OF
WOLVERHAMPTON

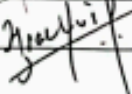
PROJECT MANAGEMENT LOG	
First Name: Ashreen	Surname: Dangol
Student Number: 2407774	Supervisor: Suraj Pandey
Project Title: EditEase	Month: December
What have you done since the last meeting	
In this previous week i completed, my flaws in the project such as video only processing one. now a whole group of videos can be processed at once, so it automatically segments and categorizes, i used streamlit to view a simple ui and to check whether everything is working or not. I also found out deepface wont work on face thats not vertically upright.	
What do you aim to complete before the next meeting	
For my next week, i would start onf frontend and backend and as well start on manuallling training ai with suitable data collection.	
Supervisor comments	
+ Complete artifacts.	

We confirm that the information given in this form is true, complete and accurate.

Date: 12/22/2025

v1

Logsheet8

Faculty of Science and Engineering School of Mathematics and Computer Science		 UNIVERSITY OF WOLVERHAMPTON
PROJECT MANAGEMENT LOG		
First Name: Ashreen	Surname: Dangol	
Student Number: 2407774	Supervisor: Suraj Pandey	
Project Title: EDITEASE	Month: December	
What have you done since the last meeting		
I set up the backend by saving generated scene data in MongoDB and building a Flask API to search and filter clips by scene type, emotion, video name, and duration. This allows video clips to be organized and retrieved programmatically without manual editing.		
What do you aim to complete before the next meeting		
In the next phase, I will continue developing the frontend with Streamlit so users can browse, filter, and preview video clips using the indexed metadata. I also plan to finish backend features like clip export and improve API stability. Additionally, I will begin manual data inspection and dataset preparation for future machine learning training, including selecting video samples and defining labeling criteria.		
Supervisor comments		
We confirm that the information given in this form is true, complete and accurate.		
Student Signature: 	Date: <u>12/29/2026</u> ⁵	
Supervisor Signature: 	Date: <u>12/29/2026</u> ⁵	

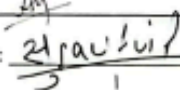
Logsheet9

Faculty of Science and Engineering
School of Mathematics and Computer Science



PROJECT MANAGEMENT LOG	
First Name: Ashreen	Surname: Dangol
Student Number: 2407774	Supervisor: Suraj Pandey
Project Title: EDITEASE	Month: December January
What have you done since the last meeting	
Since the last meeting, I integrated the Streamlit UI with the Flask API so I can search and browse scenes by scene type, emotion, duration, and video name. I completed and tested the main API endpoints (health, search, update, and export) and added FFmpeg export to cut scenes and save them into folders by scene type. I tested the full workflow by processing multiple videos, storing scene metadata in MongoDB, and checking that the UI can retrieve and display the results.	
What do you aim to complete before the next meeting	
Before the next meeting, I will finalize the data collection rules and improve the manual labeling process so manual labels and review flags are stored cleanly. I will fix a few backend issues like null emotion filtering and enforcing emotion as null when no face is detected. I will then continue reviewing and labeling scenes to start building the dataset, working toward a total of about 200 reviewed scenes.	
Supervisor comments	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 
Supervisor Signature: 

Date: 01/05/2026

Date: 01/05/2026

1

Logsheet10

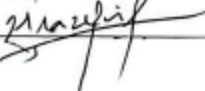
Faculty of Science and Engineering
School of Mathematics and Computer Science



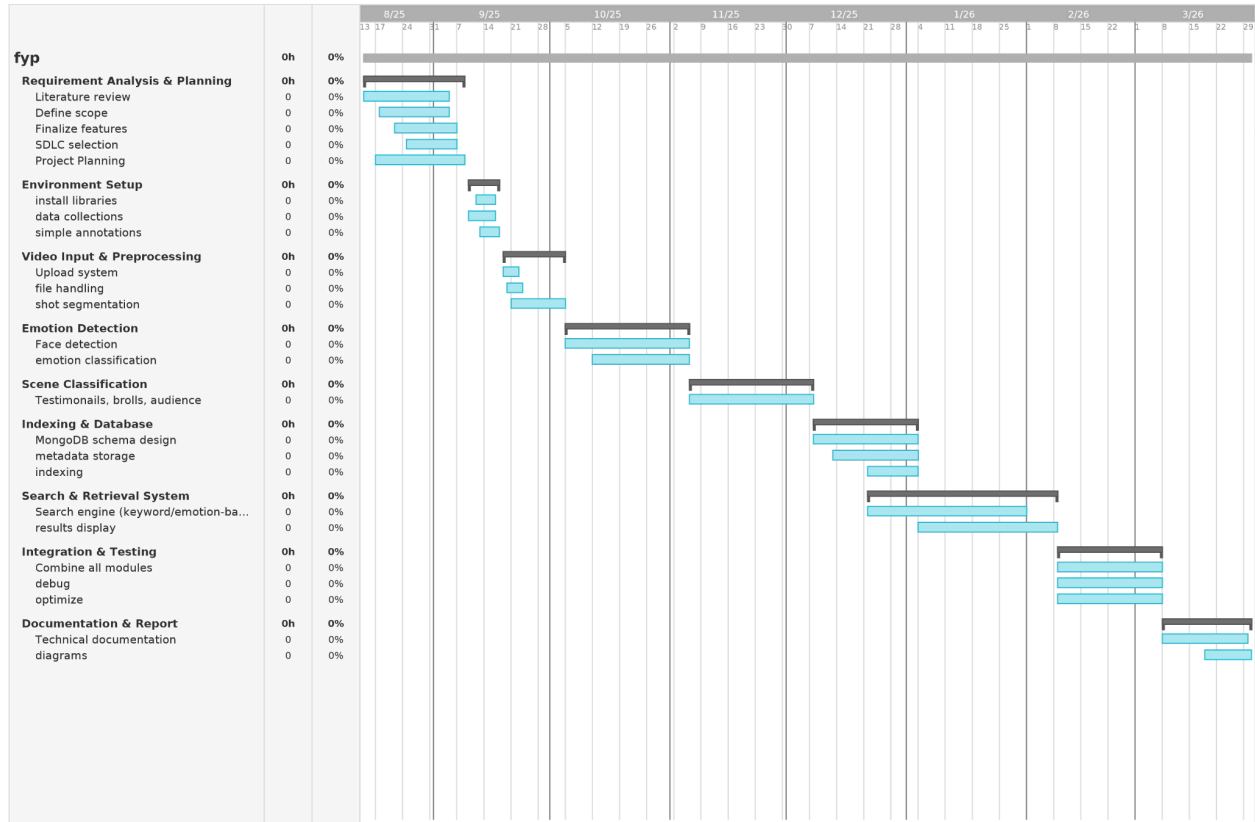
PROJECT MANAGEMENT LOG	
First Name: Ashreen	Surname: Dangol
Student Number: 2407774	Supervisor: Suraj Pandey
Project Title: EDITEASE	Month: December January
What have you done since the last meeting	
Since the last meeting, I set up the dataset-building process by defining which scenes are eligible (reviewed scenes only, excluding uncertain ones) and keeping labels consistent. I improved the backend and UI to support reviewed/uncertain/notes and ensured emotion is only stored when a face is detected. I also prepared the dataset export approach so reviewed scenes can be exported from MongoDB into a training-ready format with splits by video.	
What do you aim to complete before the next meeting	
Before the next meeting, I will complete the 200 reviewed scenes target and export a stable dataset version. I will run quality checks for missing files, wrong timestamps, duplicates, and label imbalance, and fix any issues found. I will also write up the dataset rules and prepare the Phase 10 training plan, without starting training yet.	
Supervisor comments	

We confirm that the information given in this form is true, complete and accurate.

Student Signature:  Date: 01/11/2026

Supervisor Signature:  Date: 01/11/2026

Gantt chart or development timeline



These materials provide additional evidence of the development process and support the findings presented in the report.